Deep Learning-Based HD Risk Prediction Using Sequential Neural Networks

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***Abstract*—Cardiovascular diseases (CVD) continue to be the leading cause of global death, requiring accurate and efficient prediction technology to aid clinicians in their decision-making. Conventional diagnostic methods, such as ECG or stress tests, need manual assessment and may lead to diagnostic delays or inconsistencies. Recent breakthroughs in AI and deep learning have potential for computer-automated risk prediction models that alleviate diagnostic uncertainties. This study presents a deep learning methodology (Sequential ANN) for forecasting HD. A dataset on Heart Disease (HD), containing 14 clinical factors such as sex, age, fasting blood sugar resting blood pressure, cholesterol levels and chest pain type, is employed for model training. The trained artificial neural network utilizes the Adam optimizer (AO) and binary cross-entropy loss (BCEL), leading to swift convergence and improved classification accuracy. The experimental results demonstrate that the predictive performance is outstanding, with the model attaining an overall performance of 98% (Precision: 98.01%, Recall: 98%, F1 score: 98%, ROC AUC: 97.77%). Experimental results demonstrate the effectiveness and efficiency of the proposed method for early cardiac disease prediction, applicable in real-time risk assessment and clinical decision support systems.**

Keywords*— *Heart disease (HD) prediction, Artificial Neural Network (ANN), Sequential deep learning, Cardiovascular dataset, Medical diagnosis, Early detection, Clinical decision support.

I. INTRODUCTION

Cardiovascular illnesses represent a significant global health crisis, resulting in millions of fatalities annually. Timely diagnosis and risk assessment are crucial for patients with ASUC to avert serious health repercussions and enhance their prognosis. Standard diagnostics like electrocardiograms, cholesterol profiles, and stress tests can provide significant insights into underlying issues; however, they may not necessarily present a comprehensive overview [1]. The vast volume of digital data in healthcare increasingly necessitates efficient, accurate, and automated diagnosis. Recent advancements in (AI) and (ML)have been employed as computational instruments for intricate medical datasets to reveal concealed patterns. Employing these tools within cardiovascular research could significantly enhance early prediction, diagnosis, and therapeutic planning [2].

Cardiovascular disease constitutes a significant global public health issue, resulting in millions of fatalities annually and imposing considerable financial burdens on healthcare systems. Timely identification is essential; yet, traditional diagnostic methods are often impeded by constraints in accuracy and speed, hence restricting effective intervention. The vast quantity of medical information necessitates sophisticated technologies capable of analyzing intricate patterns and making precise predictions. Artificial intelligence (AI), especially through deep learning techniques, has demonstrated significant potential in the medical industry for diagnosis by uncovering intrinsic data patterns that are not immediately observable. There is a genuine necessity for appealing and sophisticated predictive models to facilitate precise and straightforward decision-making regarding cardiac disorders, hence assisting physicians. This study aims to enhance patient outcomes and reduce mortality, advancing "Towards Data-Driven Preventive Medicine" in the context of data analytics and healthcare [3].

Several writers have investigated ML methodologies for coronary HD prediction utilizing clinical data. Logistic regression and decision trees are prevalent methodologies; yet, they often fail to model non-linear connections effectively. SVM and RF classifiers achieve superior accuracies; however, they require substantial parameter adjustment. The Naïve Bayes and KNN algorithms were also tested; despite their simplicity, they fared poorly with large datasets. Ensemble learning techniques have recently been employed to enhance predictive accuracy. DL models, such as ANN and CNN, demonstrate remarkable performance due to their ability to analyze non-linear input patterns. However, model interpretability, generalization, and implementation in real-world healthcare remain hard.

This study introduces a sequential ANN model for accurate prediction of HD based on clinical variables. The model analyzes 13 parameters, encompassing patient

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demographics, vital signs, and test findings, to identify patterns associated with cardiovascular risk. The preprocessing and normalization of the data will facilitate your model's learning from this dataset. The artificial neural network is optimized using the Adam algorithm and BCEL for effective classification. Experimental results indicate that our model attains excellent accuracy and dependability while being computationally economical enough to facilitate real-time clinical decision-making [4].

This paper's primary contributions are as follows:

- We suggested a sequential artificial neural network methodology to predict cardiovascular illness utilizing 14 clinical factors.
- Implemented preprocessing techniques to enhance data quality and model efficacy, including normalization and categorical encoding.
- Utilized the AO and BCEL to train the ANN for efficient learning and resilient classification.
- In comparison to conventional machine learning techniques, we achieved superior prediction performance with exceptional accuracy, precision, recall, and ROC AUC.
- A real-time predictive interface for probability-driven risk evaluation to support clinical decision-making in early diagnosis.

This paper is organized as follows: Section II reviews related literature, whereas Section III describes the technique. Section IV describes the experimental results, while Section V ends and suggests future research.

II. RELATED WORK

Mondal, S. A et al. [5] (2025) study introduces an ANN-based intelligent healthcare system for predicting CHD risk, utilizing advanced preprocessing and sampling techniques to address class imbalance. Experimental results indicate that attaining high validation accuracy and recall value is more desirable, suggesting the model's potential as a dependable decision support tool. Nonetheless, numerous constraints exist: reliance on a single dataset, restricted generalization, and the necessity for real-world clinical validation prior to adoption.

Lilhore, U. K et al. [6] (2025) proposed work introduces a MMAM with Deep BiLSTM (M2AM with Deep BiLSTM) to enhance HD classification through the dynamic selection of class-specific attention weights. The results demonstrate higher performance in accuracy, precision, recall, and F-measure, greatly surpassing conventional ML/DL techniques, while also reducing noise and misclassification. Nonetheless, the attention-based architecture is intricate and relies on computationally intensive preparation procedures for acquiring data representations that can be efficiently extended to our varied real-world datasets.

Kalatehjari, E et al. [7] (2024) work presents an EOA-enhanced LSTM (O-SBGC-LSTM) that effectively integrates spatial-temporal correlations for the early prevention and detection of diabetes. Experimental results indicate that the suggested model achieves superior performance, with an accuracy over 98%, surpassing conventional machine learning and hybrid DL models through the integration of fuzzy-based inference for preventive recommendations. Limitations encompass significant model complexity and the necessity for vast datasets to adequately train the systems, along with a need for further broad clinical validation proof of concepts.

Kalatehjari, E. et al. [8] (2025) study introduces a CNN-BiLSTM fusion model that utilizes both ECG and PCG inputs to improve the diagnosis and risk classification of CAD patients. The findings indicate that the bilinear fusion of CNN and BiLSTM predictions achieves a diagnosis accuracy of 97% on the PhysioNet CinC 2016 dataset, surpassing models trained with single signals or other prevalent competitors. Nonetheless, constraints encompass dependence on a singular benchmark dataset, heightened computing expenses for ensemble-based architectures, and the necessity for comprehensive real-world clinical validation.

Yu, J et . al [9] (2025) study utilizes a longitudinal deep learning model, Dynamic-DeepHit, to predict ten-year ASCVD risk by employing eight years of repeated risk factor data, rather than the conventional cross-sectional PCE approach. Outcomes the deep learning model surpassed PCE anyway. SOFA attained an AUROC of 0.815, surpassing 0.792, with comparable calibration and a net reclassification score of 0.385, highlighting the enhanced value derived from incorporating predictions of longitudinal risk factors. Caveats encompass the utilization of cohort-based data, possible computational expenses for drug development applications, and the necessity for validation in external cohorts comprising a broader patient demographic.

III. PROPOSED WORK

The aim of this study is to develop a sequential ANN-DL model capable of predicting heart illnesses with superior interpretability compared to shallow networks. The model employs 13 clinical inputs, encompassing patient demographics, vital signs, and diagnostic tests, to identify patterns potentially associated with cardiovascular disease (CVD). Data cleansing and preparation techniques, including normalization and categorical data encoding, are employed based on data quality. The ANN model comprises several hidden layers utilizing ReLU activation functions to identify non-linear correlations between dependent and independent variables. Dropout layers are employed to mitigate overfitting and enhance the model's generalization capabilities. The concluding layer is an output layer that employs sigmoid activation to assess the risk of HD. It utilizes the AO with BCEL, facilitating efficient convergence [10].

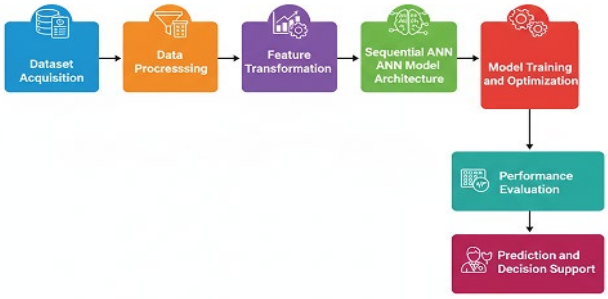


Fig 1. Block diagram of proposed model

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A. Dataset Acquisition and Understanding

This study used a cardiovascular disease dataset of 14 clinical characteristics. One aspect pertains to demographic variables such as sex and age ,while the other encompasses diagnostic factors including maximum heart rate achieved, cholesterol level, resting ECG results, resting blood pressure, exercise-induced angina, fasting blood sugar, chest pain type, slope (slope of the peak ST segment during exercise), oldpeak (ST depression induced by exercise relative to rest)and the number of major vessels [11]. The goal classes are 0 and 1, denoting the absence and presence of cardiac disease, respectively. The dataset comprises diverse category, binary, and numeric variables, rendering it suitable for (ML) and (DL) classification applications. Features are articulated in medical units, ensuring that forecasts are therapeutically relevant. This diversity of features provides a robust foundation for training models to identify patterns potentially associated with urban cardiovascular risk [12].

B. Data Preprocessing

Preprocessing was performed before the modeling phase to remove data outliers. Polygons with missing or altered values were treated accordingly to prevent bias in the learning process. Numerical properties have been normalized by min-max scaling to ensure all variables are within a uniform range. Categorical features, such as chest pain kind, resting ECG, slope of ST segment, and number of main vessels, were converted into numerical format using label encoding. Categorical variables, such as fasting blood sugar, gender, and exercise-induced angina, were maintained in their binary form. The dataset was randomly mixed to eliminate any ordering effects and thereafter divided into training and testing sets for equitable comparison. These measures were essential for maintaining a clean, consistent, and well-structured dataset for effective deep learning [13].

C. Model Architecture

The predictor is a sequential artificial neural network for binary classification. The network initiates with an input layer that processes 13 inputs supplied by the dataset. We incorporated an additional hidden layer, consisting of a fully linked layer with a ReLU activation function, enabling the learning of more intricate non-linear patterns at the input stage. Dropout layers are employed to enhance generalization and prevent overfitting. The output layer employs the sigmoid activation function to provide probabilities ranging from 0 to 1 for heart illness. The model employs an AO that determines distinct learning rates for each feature dimension based on the previous gradient, with a BCEL appropriate for binary classification. This system is intricate yet effectively balances strength and efficiency in risk distribution, ensuring consistent performance in risk type categorization [14].

D. Model Training and Optimization

The artificial neural network was trained on preprocessed data utilizing a supervised learning method. The model underwent extensive training over numerous epochs, with simultaneous modification of the corresponding weights to reduce classification error. We utilize mini-batch gradient descent to enhance model stability and decrease computational expenses. We employed the Adam method for optimization, which enhances learning rate adaptability and facilitates expedited convergence [15]. We employed BCEL during training to mitigate erroneous predictions. The entire dataset was randomly divided into training and testing subsets during the training process to evaluate performance and prevent overfitting. The accuracy and loss functions over the epochs were graphed, demonstrating that both curves converged without exhibiting signs of instability. These were several factors contributing to the success of ANN in classification [16].

E. Performance Evaluation

The model was evaluated after training with a testing dataset. To assess performance, we construct a confusion matrix comprising (FP),(TP), (FN) and (TN). Standard evaluation criteria, including accuracy, recall (sensitivity), precision, F1-score, and ROC AUC, are employed to offer a comprehensive assessment of the model's performance [17]. The model demonstrated exceptional performance across all evaluation metrics, indicating a strong capability for forecasting cardiac morbidity. Accuracy/loss curves and receiver operating characteristic (ROC) curves were employed to illustrate the learning behavior and classification efficacy. The ANN is capable of processing both linear and nonlinear data patterns, as indicated in this observation [18].

F. Prediction and Decision Support

The novel model assessed the risk of cardiovascular illness in new patients. For each input case, the model generates a likelihood score to predict cardiovascular disease (CVD). A cutoff value is established to distinguish between those with the condition and those without it [19]. The suggested methodology is demonstrated by case studies that effectively differentiate between low- and high-risk experimental units with notable accuracy and confidence. The suggested system provides a decision assistance tool for physicians that substantially decreases dispensing time while ensuring more consistent risk assessments. Employing clinical characteristics and deep learning facilitates early diagnosis for intervention, perhaps resulting in improved patient outcomes [20].

IV. RESULT ANALYSIS

The Sequential ANN model had an overall performance of 98%, indicating strong predictive capability. The confusion matrix revealed 81 TN and 114 TN, alongside two FN and three FNs, signifying minimal misclassification. Precision, recall, and F1-score were 98.01%, 98%, and 98%, respectively, indicating an effective equilibrium between sensitivity and specificity. The ROC AUC estimate of 97.77% further corroborated the model's discriminatory capacity between sick and non-diseased cases. The accuracy curves attained a convergence rate of roughly 95–98% for both training and validation, with loss values progressively declining below 0.1, signifying effective learning during information retrieval. Similarly, high confidence was

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achieved in case studies, with 99.86% for low-risk outcomes and 96.09% for high-risk outcomes.

Table 1. Sequential ANN Model Evaluation Metrics

Metric	Score (%)
ROC AUC	97.77
Accuracy	98.00
Recall	98.00
F1 Score	98.00
Precision	98.01

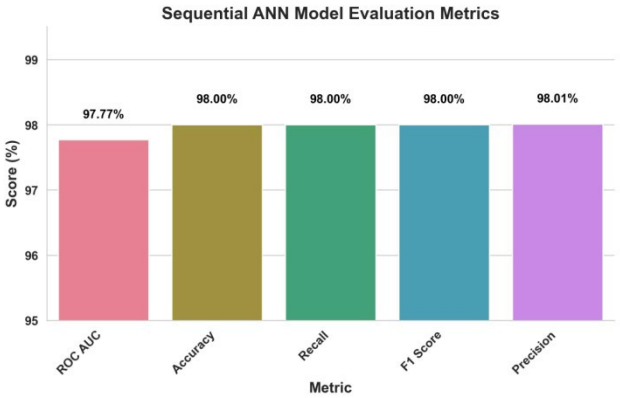


Fig 2. Sequential ANN Model Evaluation Metrics graph

Table 1 and Figure 2 show the HD prediction model's performance on numerous measures. The model's 98% accuracy proved its categorization usefulness. An effective balance between accurate positive identifications and erroneous negatives is shown by accuracy and recall scores of 98%. The harmonic mean of precision and recall, the F1-score, is 98%, indicating good predictions. The model identifies HD from non-HD with a ROC curve AUC of 97.77%. These results show that the cascaded ANN method predicts medical dangers well.

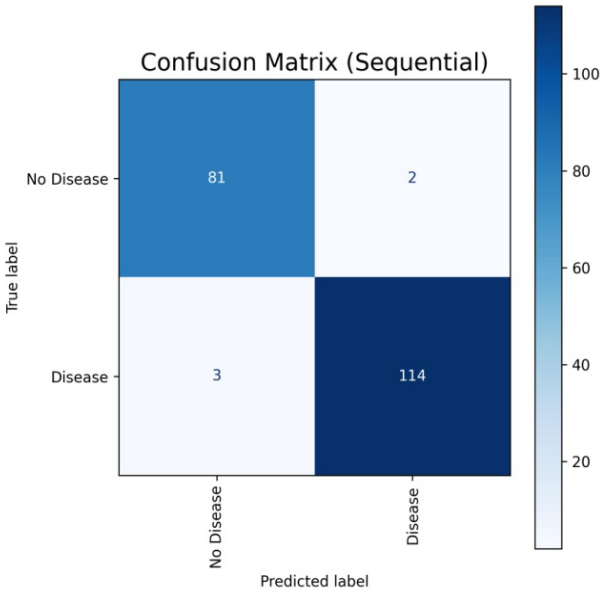


Fig 3. Confusion matrix of the Sequential ANN model

The confusion matrix of the serial ANN model depicted in Figure 3 provides a comprehensive depiction of the classification process. A total of 81 samples designated as "No Disease" were accurately identified as real negatives, whereas 114 samples classified as "Disease" were genuinely positive. Two classes were inaccurately classified as diseased (false positives), whereas three records were overlooked by the model and not identified as diseased (false negatives). The comparatively reduced misclassification rate indicates that the model is highly resilient. The matrix indicates that the model has an effective balance between sensitivity and specificity. In conclusion, it validates the efficacy of the suggested ANN in accurately detecting cardiovascular illness.

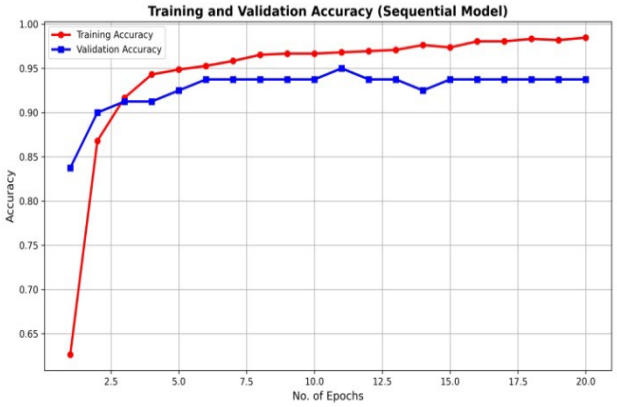


Fig 4. Training and validation accuracy of the Sequential ANN model

Figure 4 illustrates the training and validation accuracy of the Sequential ANN model over 20 epochs. The training accuracy commences at a diminished level but escalates swiftly, surpassing 95% after a few epochs. The validation accuracy also increases consistently and stabilizes at approximately 94% after several epochs. This will verify that our model can generalize effectively for the test data and assist in mitigating overfitting. Validation accuracy exhibits little indication of significant changes, as anticipated in practical datasets. The curve demonstrates effective learning and convergence, indicating that the suggested ANN model operates robustly.

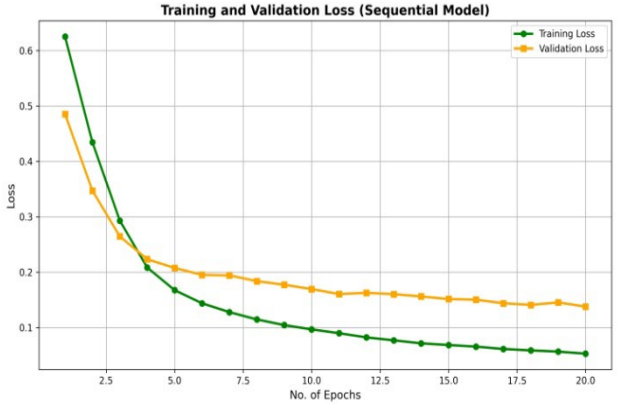


Fig 5. Training and validation Loss of the Sequential ANN model

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Figure 5 illustrates the training and validation loss of the Sequential ANN model over 20 epochs. Initially, both losses are elevated but decrease rapidly within the first few epochs (accelerated learning). The training loss steadily decreases until it gradually converges to a minimal value, signifying the effective resolution of the optimization problem. The validation loss decreases consistently, plateaus after several epochs, and remains low. The minimal disparity between the training and validation loss suggests that the model exhibits limited overfitting. Overall, the evidence supports that our ANN achieved strong convergence and reliable performance on novel data.

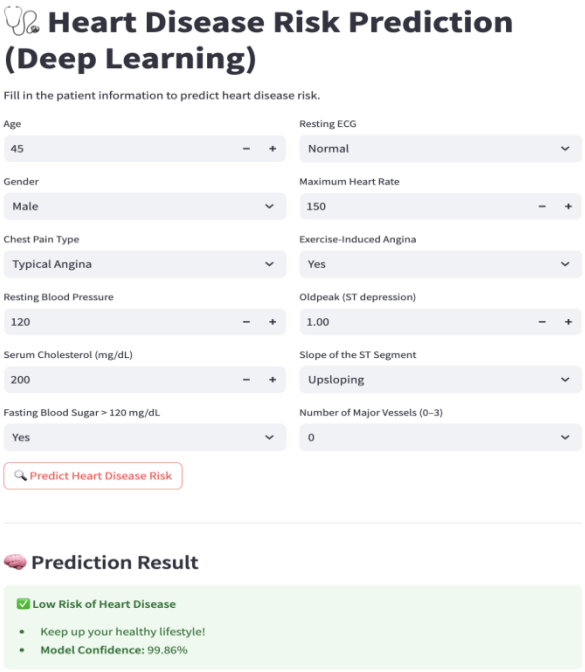


Fig 6. Low-Risk HD Prediction (Patient Input and Model Output) and

Figure 6 depicts a 45-year-old man with typical anginal symptoms, a normal resting ECG, 120 mmHg resting blood pressure, 200 mg/dL serum cholesterol, and a 150 maximal heart rate. Also considered are exercise-induced angina, fasting blood sugar exceeding 120 mg/dL, an upsloping ST segment, and no major vasculature. These clinical indicators suggest reduced HD risk, according to the model. The prediction result shows 99.86% confidence that the patient will live a healthy life. This conclusion shows that the model can confidently identify low-risk individuals.

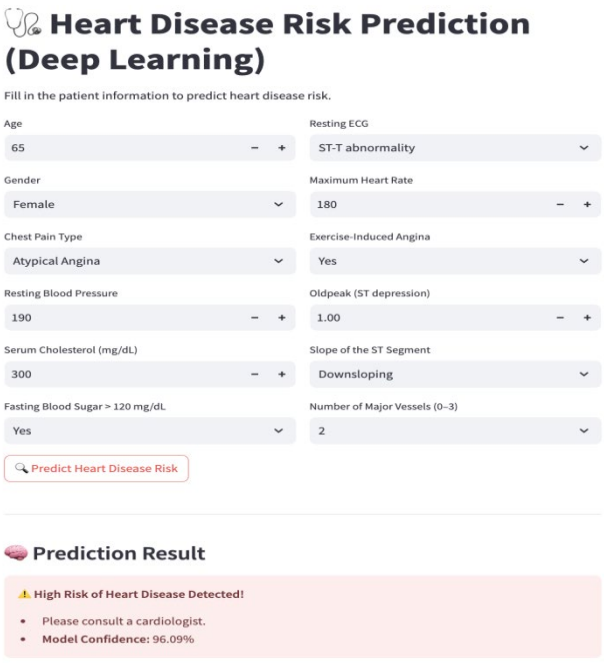


Fig 7. High-Risk HD Prediction (Patient Input and Model Output)

Refer to Fig. 7 for a 65-year-old woman with atypical angina, ST-T ECG abnormalities, a resting BP of 190 mmHg, and serum cholesterol of 300 mg/dl. Exercise-induced angina, fasting blood sugar >120 mg/dl, downsloping ST segment, and >2 major arteries are present. Maximal HR is 180. For this input, the model predicts high HD risk with 96.09% confidence. Example: The model recommends a cardiologist to the patient, identifying critical circumstances needing rapid medical expert consultation.

V. CONCLUSION

This study has demonstrated that the sequential ANN is optimal for accurately and reliably predicting the risk of HD. Utilizing 13 clinical features and extensive preprocessing, the model successfully identified both linear and nonlinear associations within the data. The performance evaluation yielded outstanding results across several criteria, hence validating the model's appropriateness for clinical applications. The case studies also validated its capacity to accurately differentiate between low- and high-risk patients. In conclusion, the suggested framework offers a pragmatic decision support instrument for early diagnosis and healthcare prevention. In the future, the model may be expanded and generalized utilizing bigger, well-diversified datasets from single-center research, employing the XAI technique in conjunction with large-scale wearable IoT devices for continuous real-time patient monitoring.

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